

CASA0006 Assessment Guidelines 2025-2026

Overview

This individual assignment aims to test your ability to conduct in-depth spatial data analysis using methods from this module. You are required to submit a single Python Notebook which contains both the code required to conduct the data analysis and accompanying text which provides context and interpretation.

Assessment Steps

1. Select a Main Dataset

You are recommended to choose one of the following datasets and define a research question that relates to **urban or spatial processes**. Feel free to use a subset of the chosen dataset, such as focusing on a specific crime type or a type of traffic incidents.

If you wish to use a different dataset, you are welcome to do so, as long as the dataset is relevant to urban or spatial processes and is appropriate for the analysis.

Dataset	Link	Description
London Crime Records	data.london.gov.uk	Crime data across London
Childhood Obesity Prevalence	data.london.gov.uk	Obesity rates by borough, ward, and MSOA
Road Safety Data (UK)	data.gov.uk	Incident location, severity, weather, road conditions, vehicle types, etc.

2. Define Your Research Question

Your research question should be specific and clear. Examples: - “What is the relationship between crime rates and local deprivation in London?” - “Is it possible to predict childhood obesity prevalence using socio-demographic variables?” - “How do road accident rates vary by location and weather conditions in the UK?”

3. Augment Your Data (Optional)

You can enhance your main dataset with additional datasets from: - Census and demographic data - Social indicators - Economic factors - Environmental variables

4. Structure Your Analysis

You need to use the [notebook template](#) for this assessment.

The analysis should be captured in a **single Python notebook** containing: - All code for data analysis - Full documentation of the analysis process - Interpretation of results - Narrative text (max 1500 words, code/comments not included)

5. Choose Your Methods

You can use *up to 4 methods* that are appropriate for your research question.

6. Include Required Notebook Sections

Your notebook must include these sections:

1. Introduction

- Include at least 3 relevant studies from credible sources (Google Scholar, CrossRef, etc.)
- Set the context for your research

2. Research Questions

- State explicitly, ending with a question mark
- Example: "What is the relationship between Covid-19 mortality rate and local deprivation in the UK?"

3. Data

- Describe your datasets
- Explain any preprocessing or cleaning

4. Methodology

- Explain the methods you chose and why
- Connect them to your research question

5. Results and Discussion

- Present findings clearly
- Interpret results in context

6. Conclusion

- Summarise key findings
- Discuss implications

7. References

- Use credible sources only

Marking Scheme

Your work will be assessed across these categories:

Category	Weight
Analysis context and research questions	15%
Data collection, handling, and presentation	15%
Correctness, depth, and scope of data analysis	35%
Visualisation	10%
Quality of writing	15%
Creativity of analytical work	10%

A more details marking scheme is below:

Criterion	A+ (80-100%)	A (70-79%)	B (60-69%)	C (50-59%)	Fail (near pass) (40-49%)	Fail (1-39%)
0 Analysis context and research questions (15%)	The report focuses on an analysis and interpretation of the dataset chosen. Each component of analysis, visualisation, database structure is detailed in the context of the overall project. The report overall conveys, excellently, the rationale of the methodology undertaken and the visualisations created compliment the story behind the data analysis extremely well. The project's background references are framed extensively to wider projects/academic literature, far	The report focuses on an analysis and interpretation of the dataset chosen. The report conveys the story behind the analysis well and the dataset chosen compliments the story and analysis undertaken. The project's background references are framed well to wider projects/academic literature, with few to no inaccurate bibliography or citations. The research question is well thought out.	The report focuses on the analysis. The wider context of the work is clearly defined and how the work places in the wider context of research. The report conveys the story behind the analysis but some flaws in execution of the analysis distract from the overall narrative of the submission. The project's background references are framed broadly to wider projects/academic literature. There may be some inaccuracies bibliography or citations. The research question is adequate and reflects some critical thinking of the topic.	The report shows a basic understanding of analysis and the data set chosen is limited in analytical scope. The story of the analysis is broken and the work raises more questions of the data than the work answers. The work satisfies the techniques that were taught during lectures but the wrong analytical techniques have been applied to the dataset. The project's background references are framed	The report shows a basic understanding of an analysis, but the data set chosen is unsuitable for an analysis. The story of the analysis is broken and incoherent. The work raises more questions of the data than the work answers. The report details the wrong techniques to analyse the data and provides no insight to the data being analysed. Very little background research is evident beyond use of the datasets. Bibliography	A poor attempt at the problem using an unsuitable dataset. The story behind the analysis is not present and the report has been rushed and thrown together haphazardly. There is no evidence of wider background research at all. No research questions.

Criterion	A+ (80-100%)	A (70-79%)	B (60-69%)	C (50-59%)	Fail (near pass) (40-49%)	Fail (1-39%)
	beyond that on the prescribed reading list. The research question is very interesting and insightful.			but may be limited to the prescribed reading list/datasets. There may be obvious inaccuracies bibliography or citations. A fair research question is proposed but is not interesting.	may be incomplete or inconsistent and citations poor. A research question is proposed but needs substantial improvements.	
1 Data collection handling, and presentation (15%)	A detailed description of the data is present. Data has been extensively treated and justified in preparation for analysis. There is a table that describes the variables selected for analysis so that the readers can easily understand the variables used for analysis.	A detailed description of the data is present. Data has been extensively treated and justified in preparation for analysis. There is a table that describes the variables selected for analysis.	A detailed description of the data is present. Data has been treated and justified in preparation for analysis. There are no obvious errors in variable selection.	A detailed description of the data is present, including data provider, country, and URL (if the URL exists). Data has been treated in preparation for analysis. There are obvious errors in variable selection, for example, including identification columns for analysis, or incorrectly treating categorical variable as numerical ones.	There is some description of the dataset but essential information is missing, such as provider, country, or URL (if the URL exists). Data is incomplete and is not compatible for data analysis. The data is unstructured and no obvious work has been carried out on the dataset.	No data source has been identified or collected, or data source incomplete and unavailable

		Fail (near pass) (40-49%)					Fail (1-39%)
Criterion	A+ (80-100%)	A (70-79%)	B (60-69%)	C (50-59%)			
2 Correctness, depth and scope of data analysis (35%)	The work shows extensive knowledge of the topic chosen and an in-depth understanding of the links between the problem, data and method(s) employed to solve it. Shows considerable insight into any shortcomings of the analysis and demonstrates critical reflection on the finished product.	The work shows good knowledge of the chosen topic and a clear appreciation of the links between the problem, data and method(s) employed to solve it. Critical appraisal of the finished product is thoughtful and relevant. No errors in the notebook.	The work shows good knowledge of the chosen topic although there may be some minor issues with the choice of data and/or methods at the lower end of the scale. At the upper end of the scale, matching data and/or method to the problem is sound. Some evidence of critical reflection is displayed. Some minor errors in the code.	The work shows a dequate knowledge of the chosen topic but choice of data and/or methods could have been better. Critical reflection may be poor, or lacking entirely at the lower end of the scale. There are a few errors in the code but they don't impact the presentation of the work.	The work shows poor knowledge of the chosen topic and the choice of data and/or methods is ill logical. Critical reflection is an alien concept. There are existing errors in the code which prevent the code from executing, or the results of coding are largely different from the discussion in the text.		The work shows that methods taught were not understood at all.
3 Visualisation (10%)	Figures used to convey the visualisation of data analysed are of publishable quality; they are clear, well labelled and convey the intended information expertly.	Figures used to convey the visualisation of data analysed are excellent; they are clear, well labelled and wholly appropriate. Visuals complement the report and display an understanding of the methodology executed within the report.	Figures used to convey the visualisation of data analysed are overall good and well chosen, but at the lower end of the scale may contain minor errors. Visuals are lacking in clarity.	Figures used to convey the visualisation of data analysed are adequate but errors detract from their usefulness. Visuals are lacking in context. The graphics may not be referenced within the document.	Figures used to convey the visualisation of data analysed are poor and do not aid understanding in any way. The graphics are not referenced within the document and are out of order/place.		Figures used to convey the visualisation of data analysed are bad or missing entirely.
4 Quality of writing (15%)	The final piece of work is presented to a professional standard. Exceptionally well written; stylish with no errors in	The final piece of work is presented very well and would only require minor editing. Very well written with	A good, well written piece of work with few errors in spelling, punctuation or grammar. The work shows	The final piece of work may lack polish and would need attention. A more-or-less	The final piece is poorly presented and would need serious editing. A more-or-less		The final piece is very badly presented. Only a complete re-write would

Criterion	A+ (80-100%)	A (70-79%)	B (60-69%)	C (50-59%)	Fail (near pass) (40-49%)	Fail (1-39%)
	spelling, punctuation or grammar. The piece of work is clearly and logically structured and enjoyable to read.	virtually no errors in spelling, punctuation or grammar. The piece of work is clearly and logically structured and flows well.	structure and is clearly presented. The work may require the odd edit.	competent piece of work but may contain some errors in spelling, punctuation or grammar. The may lack structure and presentation could be improved.	weak piece of work containing a number of errors in spelling, punctuation or grammar. The piece may lack any kind of structure.	bring it up to standard. A poor piece of work. Written English is bad, with numerous errors in spelling, punctuation and grammar.
5 Creativity of analytical work (10%)	The work is outstandingly creative and approaches the problem within a clear and analytical mind set. The execution of the analysis is also excellent and shows meticulous planning and understanding to the problem being addressed. The execution of the analysis is sound and the student has demonstrated understanding to the analytical process. At each stage the narrative is inferred by the analysis of the previous stage and the analysis has been planned from the outset.	The work is creative in the execution and shows that the student has thoroughly about the problem being tackled. At each stage the narrative is inferred by the analysis of the previous stage and the analysis has been planned from the outset.	The work is creative in nature, both in the execution and the thought process is apparent within the narrative. The execution is sound and the student has demonstrated understanding to the analytical process.	The work shows some creativity which is satisfactory, but some flaws in the execution or assertions are made within the work.	The work shows little to no creativity but falls short in the execution and understanding.	The work shows no creativity or relevance in research questions or methods, and there are major problems in the execution.

Key Considerations

1. Use Rate, Not Count

In most analysis, **rate is more appropriate than count** because it normalises data by accounting for population size or scale.

Example: In road safety research, calculate the rate of traffic incidents by normalising the count of incidents by traffic volume or population size.

2. Use Credible References

- References must be relevant and credible (Google Scholar, CrossRef)
- **Do not use** unverified references generated by LLM - this is BAD ACADEMIC PRACTICE and will be penalised

3. Choose Methods Wisely

- Ensure all methods are directly relevant to your research question
- Avoid including irrelevant or excessive methods; this reduces submission quality

Major Problems to Avoid

A high-quality submission (distinction/merit level) should NOT contain these issues:

1. **✗ ID Columns** - Including identification columns for analysis without justification
2. **✗ Categorical as Numerical** - Treating categorical variables as numerical ones incorrectly
3. **✗ Code-Discussion Mismatch** - Inconsistency between code results and discussion (e.g. model accuracy differs)
4. **✗ Code Errors** - Code in the notebook is not run or contains errors
5. **✗ Notebook-PDF Mismatch** - Python notebook and PDF file are largely different
6. **✗ Excess printing dataframe** - Printing large dataframes frequently or after each step is not necessary

Submission

The deadline for the assessment is Monday, 27 April 2026 @ 10:00.

What to Submit

Part 1: Python notebook (or .zip file with notebook + dataset files)

Part 2: PDF file exported from the notebook

Submit **both parts separately** in two tabs on Moodle. The submission timestamp is based on the later of the two parts.

Critical Rules

You will receive a mark of 0 if:

- You fail to submit either Part 1 or Part 2
- The content of Part 2 is largely different from Part 1

Dataset Requirements

- Share your dataset in a **GitHub repository** and remotely read it in the notebook (e.g. using `read_csv`)
- If data size exceeds 100 MB (GitHub limit), submit a .zip file with the notebook and data

Library Requirements

- If possible, use only libraries from the **recommended computing environment** (Podman/Docker/Anaconda)
- If you must use other libraries (including fastai):
 - Clearly state the library names and version numbers

Execution Requirements

- **Notebook must execute within 1 hour** at submission; please remove unnecessary coding from the notebook
- Use Jupyter's 'Restart & Rerun all' before submission to verify viability
- If data preprocessing requires significant time, you may provide preprocessed data with a detailed description

Warning

Penalty will apply if the code is not run or results contain errors or are not clearly presented.

Exporting to PDF

Steps to Export

1. In the web browser running your notebook, right-click
2. Select '**Print**'
3. In the printer dialog, select '**Save as PDF**'
4. Choose your save location

Alternative methods are acceptable as long as the PDF is **text-selectable**.

Moodle Warning

You may see a warning about supported file types when uploading your notebook (see below). You can **safely ignore this**.

You must upload a supported file type for this assignment.

Accepted file types are; .doc, .docx, .ppt, .pptx, .pps, .ppsx, .pdf, .txt, .htm, .html, .hwp, .odt, .wpd, .ps and .rtf

Checklist before submission

Before submitting, ensure:

- ☐ Your notebook runs completely in under 1 hour. Don't include lots of data pre-processing in the notebook.
- ☐ You've run 'Restart & Rerun all' and all cells execute successfully
- ☐ Your research question is clearly stated
- ☐ All methods are appropriate for your research question
- ☐ Your visualisations are clear and meaningful
- ☐ Your writing is clear and accessible
- ☐ Your PDF is exported and text-selectable
- ☐ You're submitting both notebook and PDF to Moodle
- ☐ Your notebook and PDF contain consistent content

